

Data Driven Systems

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Lab Assignment: a brief introduction to spatial data systems

Part 1

The GIS objects supported by PostGIS are a superset of the "Simple Features" defined by the OpenGIS Consortium (OGC). The OpenGIS specification defines standard ways to define spatial objects, namely the Well-Known Text (WKT) form and the Well-Known Binary (WKB) form. Both WKT and WKB include information about the type of the object and the coordinates that compose the object.

The following lists some examples of text representations (WKT) of spatial objects:

- POINT(0 0)
- LINESTRING(0 0,1 1,1 2)
- POLYGON((0 0,4 0,4 4,0 4,0 0),(1 1, 2 1, 2 2, 1 2,1 1))
- MULTIPOINT(0 0,1 2)

The OpenGIS specification also requires that the internal storage format of spatial objects includes a spatial referencing system identifier (SRID) when dealing with geographical coordinates. The SRID is required when creating spatial objects for insertion into the database.

Docs: http://postgis.net/docs/manual-2.1/using_postgis_dbmanagement.html#RefObject

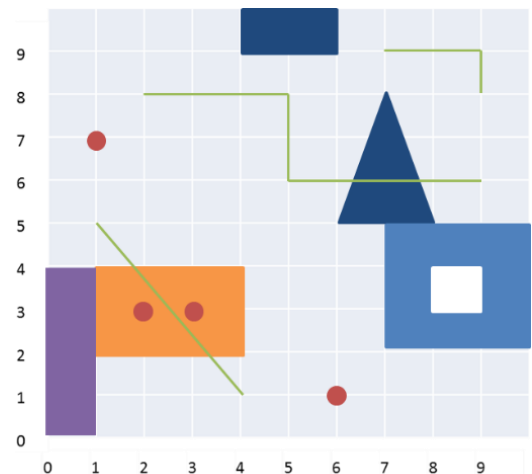
Run PGAdmin, login as *postgres* and execute the following requests.

1. Create a spatial database using the template *postgis_xx_sample*.
2. Execute and interpret the result of the following queries.

```
SELECT ST_GeomFromText('POINT(48.998832 12.093085)', 4326);
```

```
SELECT ST_asText(ST_GeomFromText('POINT(48.998832 12.093085)', 4326));
```

3. Create a table with a geometry column and insert the spatial object displayed in the following figure.
4. List all records in the table and display all geometries.
5. Run QGIS desktop application and add a PostGIS layer to display the objects in the table just created.



Part 2

Geofabrik (<http://www.geofabrik.de/>) is a website that provides free, community-maintained geospatial data, such as that produced by the OpenStreetMap (OSM) project. It offers a compelling alternative to the data provided by major commercial players in the geospatial industry. Country-specific datasets can be downloaded from <http://download.geofabrik.de/>. The description of the datasets is available at <http://www.geofabrik.de/data/geofabrik-osm-gis-standard-0.3.pdf>.

1. Download the maps of Portugal in shapefile format (portugal-latest-free.shp.zip) from the Geofabrik website.
2. Create a project in QGIS and display a map with 2–3 layers, for example roads, water lines, railways, or points of interest. To add a shapefile in QGIS, use the Add Vector Layer command.
3. Import the shapefile of Portugal's roads into PostgreSQL using, for example, shp2pgsql.
4. How many kilometers of cycling paths exist in Portugal?
5. How many hospital infrastructures are there in Portugal, and what is their total area?